

Track 46: ANSYS Simulation & Analysis

June 2026

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Tier 3 — Specialized | Departments: Mechanical · Civil · Mechatronics · EEE

Skill path: FEA Concepts → Static Structural → Modal → Thermal → CFD Intro → Explicit Dynamics → Validation

Related: [Track 52 — SolidWorks](#) · [Track 36 — BIM/Structural](#)

Track Overview

Who this is for: Mechanical, Civil (structural), Mechatronics, and EEE students needing FEA/CFD simulation skills for design validation.

Prerequisites: Strength of materials, basic CAD ([Track 52](#) or [35 AutoCAD](#)).

Tools: ANSYS Mechanical, ANSYS Fluent (intro), SpaceClaim/DesignModeler, student license.

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	FEA awareness	Static analysis report
1 Month	80–100	Junior analyst	Modal + thermal study
3 Months	250–300	Simulation engineer intern	Validated design optimization
6 Months	500–600	CAE analyst portfolio	Multi-physics project

2-Week Crash Course

Days	Topics	Deliverable
D1–2	FEA theory, meshing, boundary conditions	Meshed simple part
D3–4	Static structural analysis, stress/strain	FEA report with FOS
D5–6	Modal analysis, natural frequencies	Vibration report
D7–8	Thermal analysis intro	Heat transfer simulation
D9–10	Capstone	Simulation validation report

1-Month Foundation

Week	Topics	Project
W1	Mesh quality, convergence, material models	Benchmark validation
W2	Contact, bolts, joints	Assembly FEA
W3	Fatigue analysis intro	Life prediction study
W4	CFD basics (Fluent) OR explicit dynamics intro	Multi-analysis project

Department-Specific Capstones (3-Month)

Department	Capstone Example
Mechanical	Optimized bracket/gear stress analysis
Civil	Structural beam/column FEA validation
Mechatronics	Robot arm structural analysis
EEE	Thermal analysis of motor/inverter housing

Assessment Rubric

Weekly Quiz 20% | Simulation labs 40% | Project 30% | Portfolio 10%

Appendix: Mentor Notes

Tier 3 **ANSYS** course. Requires CAD models from SolidWorks/CATIA tracks or provided geometry.